

KARTHIK NAIR

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SUMMARY

Computer Science undergraduate with interests in cloud computing, distributed systems, backend engineering, and AI-assisted software development. Experienced in building end-to-end software solutions across web, mobile, and cloud-native platforms involving backend services, databases, authentication, deployment, infrastructure integration, and real-time systems. Co-author of a published research paper and two-time international hackathon finalist, with hands-on experience delivering scalable, AI-powered, and production-oriented software solutions.

EDUCATION

B.Tech, Computer Science and Engineering

Aug 2023 – May 2027

SRM Institute of Science and Technology, Ramapuram — CGPA: 9.59 / 10

STANDARDIZED TEST SCORES

IELTS Academic — Overall Band 7.0 (CEFR C1)

Reading: 8.5 | Listening: 7.0 | Writing: 6.5 | Speaking: 6.5

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, SQL, HTML/CSS, C, C++

Frontend: React, Next.js, Tailwind CSS, Firebase Authentication

Mobile Development: Android Development (Kotlin), Jetpack Compose

Backend: Node.js, Express.js, REST APIs, Authentication & RBAC, JWT, NextAuth.js, Socket.IO, File Upload Pipelines, Payment Integrations, API Design, FastAPI

Databases: MongoDB, MySQL, PostgreSQL, Supabase

Cloud & Infrastructure: Docker, Google Cloud Platform (GCP), AWS S3, Linux, SSH, Nginx, Vercel, Containerization

AI & Automation: Gemini API, Prompt Engineering, OCR Pipelines, OpenCV, ML Kit, LLM Integration

Web3: Solidity, Smart Contracts, Web3.js, Ethers.js, Hardhat

Developer Tools: Git, GitHub, Postman, VS Code, FFmpeg, Figma, Android Studio

Core CS / Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Object-Oriented Programming (OOP), System Design, Database Management Systems, Software Engineering

EXPERIENCE

Backend Developer (Part-Time) | Quantum Scripts

Nov 2025 – Jan 2026

- Migrated the core data layer from Supabase to MongoDB, giving the payments and credits system transactional control it previously lacked and eliminating a source of data-consistency risk in production.
- Engineered REST APIs handling large video uploads end-to-end — ingestion, asynchronous processing pipeline, and retrieval — for an AI-driven subtitle-generation SaaS product.
- Owned Docker containerization of backend services, including container lifecycle management on remote cloud instances via SSH, to standardize deployment across environments.
- Integrated AWS S3 for large media file storage and built backend logic to reliably persist Razorpay payment transactions and usage credits.
- Debugged and supported live deployments on GCP, resolving production issues affecting active users.

Frontend Developer, Industrial Training | Integral Coach Factory (ICF), Ministry of Railways

Jun 2025 – Jul 2025

- Developed a complaint management portal using Next.js, React, and Tailwind CSS during a 15-day industrial training program at the ICF IT Centre.
- Built responsive user interfaces and implemented routing workflows aligned with organizational requirements for internal complaint handling.

Technical Member | Codekrafters

Nov 2024 – Present

- Collaborate on full-stack and blockchain application prototypes, contributing to hackathon projects involving frontend development, wallet integration, and smart contract interactions.
- Participate in peer reviews, debugging, workshops, and community-led technical initiatives, expanding practical experience in software engineering and Web3 development.

PROJECTS

ProjectForge — AI-Powered Proof-of-Work & Accountability Platform

Next.js • Firebase • MongoDB • Vercel

- Architected a full-stack accountability platform for goal tracking and milestone verification, built around tamper-evident proof management — SHA-256 file hashing, append-only audit trails, and secure multi-format uploads — so evidence cannot be silently altered after submission.
- Integrated AI-powered evidence analysis to auto-generate verification summaries and confidence scores while keeping the final decision in a human-in-the-loop workflow.
- Implemented Firebase Authentication with role-based access control (RBAC) to enforce multi-user data isolation and administrative oversight, deployed globally on Vercel with MongoDB-backed persistence.

SmartSpend — AI-Powered Personal Finance Platform

Next.js • TypeScript • MySQL (TiDB Cloud) • Gemini API

- Built a full-stack fintech platform for expense tracking, budgeting, and multi-currency financial planning, with secure authentication via NextAuth.
- Designed an OCR-based receipt and bank-statement pipeline that extracts, validates, and reconciles transactions automatically, feeding a real-time reporting dashboard.
- Combined the Gemini API with rule-based logic to auto-categorize spending and surface personalized financial insights.

SentinelRoute — Operational Logistics Intelligence Platform

Next.js • TypeScript • MongoDB • Firebase • Socket.IO

- Built a multi-tenant operational logistics platform with server-authoritative tenant isolation, secure company onboarding, role-based access control, immutable audit trails, and an interactive Command Action Panel for enterprise operations.
- Engineered a real-time, event-driven architecture using Socket.IO with centralized event dispatching, shared state synchronization, company/entity rooms, presence tracking, and live updates across operational dashboards, command centres, and shipment collaboration.
- Developed an explainable operational intelligence engine combining deterministic risk scoring, regional weather/traffic analysis, cargo sensitivity, urgency, and rule-based recommendations with Gemini-powered route explanations, confidence scoring, operational impact metrics, and trade-off analysis for command-centre decision support.

Trust Buddy — AI Financial Security Assistant

Kotlin • Jetpack Compose • FastAPI • Gemini • Groq • ML Kit

- Developed an AI-assisted financial security application that evaluates suspicious messages, URLs, QR codes, and payment-related content using deterministic risk scoring combined with LLM-generated explanations.
- Integrated Android-based OCR, QR scanning, voice input, and a FastAPI backend to analyze security signals such as phishing attempts, brand impersonation, credential harvesting, and malicious links.
- Implemented a RAG-enhanced assessment workflow that delivers risk classifications, threat indicators, and actionable safety recommendations while ensuring AI explanations cannot override security decisions.

PUBLICATIONS

- Co-authored and presented "SmartSpend: A Goal-Based Personal Financial Planning Platform for Awareness-Driven Savings," published in the proceedings of ICGMRFT 2026.

ACHIEVEMENTS

- IEEE YESIST12 2026 — International Finalist, Maker Fair track, selected from 2,000+ submissions worldwide, for SmartSpend.
- USAII® Global AI Hackathon 2026 — International Finalist recognition, for ProjectForge.
- Aloha Abacus 2019 — National Level 1st Runner-Up and State Level Winner.
- Aloha Abacus 2012 — State Level Winner (Tiny Tots Category).

LANGUAGES

English • Tamil • Malayalam • Hindi • Sanskrit • German